Team NeuroBridge

PROPOSED SOLUTION: NEUROBRIDGE PLATFORM

HACK2HEAL
2.0 • 2026

Bidirectional Double Empathy Translation & Wearer-Controlled Multimodal Healthcare Bridge

TRADITIONAL AAC APPROACH (Deficit & Masking)

✕ Unilateral Masking Focus: Trains autistic users to suppress natural traits and mimic neurotypical norms.
✕ Prohibitive Hardware Cost: Dedicated AAC devices (Dynavox) cost $4,000–$8,000 with long insurance delays.
✕ One-Way Voice Output: Static symbol grids speak out loud but offer zero subtext or intent clarification.
✕ Blind to Situational Mutism: Assumes motor ability to navigate menus during acute sensory shutdowns.
✕ Naive AI Abuse Gaslighting: Generic chatbots mistake hostile bullying ('nobody likes you') for friendly idioms.

NEUROBRIDGE DUAL-ENGINE (Double Empathy AI)

✓ Reciprocal Subtext Translation: Translates idioms for autistic wearers and directness for neurotypical peers.
✓ Ubiquitous Zero-Cost Hardware: Runs in browser on existing laptops/phones (https://hack2health.vercel.app).
✓ 4-in-1 Multimodal Timeline: Unifies Speech, Switch-Scanning, AAC Chips, and 3D MediaPipe Stim Gestures.
✓ Wearer-Sovereign Safety Net: On-demand GPS & Communication Passport dispatch without 24/7 surveillance.
✓ Anti-Gaslighting Hostility Shield: Detects malicious insults and protects against Rejection Dysphoria (RSD).

THE PROBLEM WE SOLVE (The Social Cliff)

The 40% Segregation Cliff: Over 40% of autistic students spend school in segregated classes (NCES Data).
Abrupt Aid Termination: Transition to college/work brings sudden loss of 1:1 paraprofessional aides.
Situational Mutism In Crises: High sensory environments (cafeterias >75dB) cause vocal speech loss.
Masking & Autistic Burnout: Forcing neurotypical imitation causes chronic fatigue, anxiety, and depression.
Dignified Healthcare Mission: Empowering autonomous social agency and mutual cross-neurotype respect.

OUR MULTIMODAL ASSISTIVE INTELLIGENCE LOOP

1. **INGEST** — Audio, Switch, AAC, 3D Stims
2. **RADAR** — Acoustic dB & Noise Monitor
3. **KINEMATICS** — 800ms Buffer & Fingertip Curl
4. **DOUBLE EMPATHY** — Milton (2012) Pragmatic Rules
5. **DUAL-ENGINE** — Gemini 2.5 Flash + Offline Regex
6. **CLARIFY** — Subtext Cards & Intent Shield
7. **SAFETY NET** — GPS Beacon & Medical Passport
8. **DELIVER** — SSE Dashboard & Peer Circles

⚡ ADAPTIVE DUAL-ENGINE: Continuously personalizes vocabulary & suppresses downvoted idioms in real time

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TECHNICAL APPROACH & SYSTEM ARCHITECTURE

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DATA SOURCES

Spoken Audio
Web Speech API streaming raw acoustic input in browser

Switch Scanning
Single-switch scanning keyboard for fine motor differences

Typed AAC Chips
Structured quick-response non-verbal symbol cards

3D Hand Tracking
MediaPipe 21-joint 3D kinematic landmark coordinates

Sensory Decibels
Web Audio API real-time acoustic sound pressure (SPL)

DATA PROCESSING

Input Ingestion
Unified stream normalization via centralized /api/analyze

Kinematic Buffer
12-frame rolling buffer (~800ms) tracking velocity & reversals

Curl Verification
Fingertip-wrist angle check eliminates resting hand false alarms

Data Cleansing
Disfluency & filler word extraction preserves core sentiment

Session State
Local cache persists calibrated gestures & emergency profile

PREDICTION ENGINE

Google Gemini 2.5
Primary LLM: deep pragmatic subtext resolution & reasoning

Offline Heuristics
Zero-downtime regex engine for transit tunnels / basements

Double Empathy
Bidirectional classification: idiom, sarcasm, bluntness, hostile

Anti-Gaslighting
Shield: Detects malice, refuses to excuse bullying as idioms

Personalization
Real-time feedback loop suppresses disliked clarification types

SOVEREIGN OUTPUTS

Clarification Cards
Real-time contextual meaning displayed in chat timeline

Sovereign Safety Net
GPS emergency beacon & situational mutism alert dispatch

Communication ID
Digital medical card for first responders & educators

Tactful Exit Card
One-tap polite sensory recharge boundary notifications

Peer Social Hub
Parallel-play virtual co-working with zero small-talk

TECHNOLOGIES USED

Frontend Core
React 19, Vite 8.3, Glassmorphism CSS Design

Backend Engine
Node.js, Express.js REST API, Server-Sent Events

AI & Vision Core
Google Gemini 2.5 Flash, MediaPipe Vision Wasm

Accessibility APIs
Web Speech API, DeviceMotion (G-Force), Geolocation

Deployment
Vercel Edge Serverless, GitHub CI/CD, Automated Tests

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FEASIBILITY AND VIABILITY ANALYSIS

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✓ FEASIBILITY

• Zero Hardware Barrier
Runs on ubiquitous web browsers (Chrome, Safari, Edge) on existing student phones and laptops. Eliminates $7,000 proprietary hardware barriers.

• Production Proven Today
Live full-stack deployment active on Vercel at https://hack2health.vercel.app. Instant edge execution across worldwide edge nodes.

• Cost & Scale Efficiency
Serverless micro-architecture executes Gemini 2.5 Flash only on demand, costing <$0.01 per conversation with zero idle infrastructure cost.

⚠ KEY CHALLENGES

• High Acoustic Noise
Cafeteria background chatter (75–85 dB) degrades standard browser microphone speech recognition accuracy during social transitions.

• Mobile OS Sensor Throttling
Mobile Android and iOS operating systems suspend background browser tab sensors and camera feeds when the screen is locked.

• AAC Typo & Lexical Noise
Autistic disfluency, dyspraxia, or rapid typing generates phonetic spelling errors ('neeeed a brek') that bypass simple regex matching.

OUR STRATEGY

• Multimodal Input Fallback
When noise peaks on the Sensory Radar, the wearer seamlessly shifts to single-switch scanning or one-tap quick-response AAC chips with zero downtime.

• Native Daemon Packaging
Phase 3 roadmap packages the web app into a Capacitor/React Native daemon with an Android Foreground Service for locked-screen fall monitoring.

• LLM Intent Reconstruction
Gemini 2.5 Flash semantically extracts the core emotional/functional intent from disjointed spelling without judging grammar or penalizing the user.

IMPACT AND BENEFITS ASSESSMENT

POTENTIAL IMPACT ON TARGET AUDIENCE

AUTISTIC YOUNG ADULTS (14-24)

Overcoming The Social Cliff

- Eliminates reliance on 1:1 human aides upon graduation.
- Prevents situational mutism panic via quick AAC & stims.
- Tactful Exit Cards lower autistic masking burnout.
- Empowers genuine self-advocacy in social spaces.

PARENTS & GUARDIANS

Sovereign Peace of Mind

- Real-time GPS safe perimeter tracking on Guardian View.
- Alerts sent automatically during acute sensory distress.
- Zero invasive 24/7 spying: telemetry sent by wearer consent.
- Direct emergency calling with real-time location details.

EDUCATORS & PEERS

Mutual Understanding (Double Empathy)

- Educates peers: bluntness is clear honesty, not malice.
- Explains stimming as joyful energy regulation, not crisis.
- Equips college counselors with Communication Passports.
- Promotes inclusive parallel-play study environments.

✓ BENEFITS OF THE SOLUTION

SOCIAL BENEFITS

Inclusion Without Forced Masking

- Fosters authentic neurodivergent friendships.
- Social Wingman matches intense shared passions.
- Replaces anxiety with structured, predictable interactions.
- Destigmatizes stimming and AAC communication in public.

ECONOMIC BENEFITS

Democratizing $7,000 Medical Tech

- Saves families $4,000–$8,000 per user vs proprietary hardware.
- Reduces institutional special education aide expenses.
- Zero subscription barrier: free open-access web prototype.
- Runs on existing hardware without insurance delays.

HEALTHCARE & CLINICAL BENEFITS

Preventative Mental Health

- Drastically lowers autistic burnout and clinical depression.
- Reduces emergency room visits from panic & disorientations.
- First responders receive immediate sensory crisis protocols.
- 18 automated self-tests guarantee verified API robustness.

RESEARCH AND REFERENCES

PEER-REVIEWED CLINICAL & SCIENTIFIC LITERATURE

1. The Double Empathy Problem (Milton, 2012)

Milton, D. E. (2012). 'On the ontological status of autism: the 'double empathy problem'.' Disability & Society, 27(6), 883-887.
Link: https://doi.org/10.1080/09687599.2012.710008
Impact on Solution: Theoretical foundation: proves communication mismatch is reciprocal, not an autistic deficit.

2. Educational Segregation & The Social Cliff (NCES, 2024)

U.S. Dept. of Education, National Center for Education Statistics. Digest of Education Statistics, Table 204.60 & IDEA Section 618.
Link: https://nces.ed.gov/programs/digest/d23/tables/dt23_204.60.asp
Impact on Solution: Sourced data: Over 40% of autistic students are segregated in K-12 schooling, facing the Social Cliff upon transition.

3. Defining Autistic Burnout & Camouflaging (Raymaker et al., 2020)

Raymaker, D. M., et al. (2020). 'Having All of Your Internal Resources Exhausted: Defining Autistic Burnout.' Autism in Adulthood, 2(2), 132-143.
Link: https://doi.org/10.1089/aut.2019.0079
Impact on Solution: Clinical basis for Tactful Exit Cards: continuous masking leads to severe cognitive and functional exhaustion.

4. Rejection Sensitive Dysphoria & Neurodivergence

Bedrossian, L. (2021). 'Understand and address rejection sensitive dysphoria.' Mental Health Practice, 24(6).
Link: https://doi.org/10.7748/mhp.2021.e1564
Impact on Solution: Guides our Anti-Gaslighting Hostility Shield: protects wearers from misinterpreting abusive insults as benign idioms.

LIVE PLATFORM, OPEN-SOURCE CODE & SYSTEM ASSETS

- **Live Deployed Application**
  https://hack2health.vercel.app
  Active Vercel production deployment with Google Gemini 2.5 Flash LLM integration.

- **Mobile Guardian Safety Companion**
  https://hack2health.vercel.app/?view=guardian
  Live SSE real-time telemetry dashboard streaming GPS coordinates & decibel readings.

- **Official GitHub Repository**
  https://github.com/tharunya0105-afk/hack2health
  Full open-source codebase with React 19 frontend, Express backend & gesture modules.

- **Automated End-to-End Test Suite**
  backend/test-suite.js
  18 automated tests verifying multimodal ingestion, scanning keyboards & adversarial speech.

- **Complete Video Demonstration**
  gallery/neurobridge_demo_video.mp4
  1.63-minute 30FPS MP4 video walkthrough covering all 15 system capabilities.

- **MediaPipe HandLandmarker API**
  https://developers.google.com/mediapipe/solutions/vision/hand_landmarker
  Float16 WebAssembly hand tracking tracking 21 3D landmarks in-browser.